

Soft Spaces - Phase 3.2

Checkpoint: Global $\pm E$ Symmetry and Local Hotspot Mechanism

Working checkpoint - 10 September 2026

Purpose. This checkpoint records the current Phase 3.2 result chain before further mechanism work. It separates what is numerically established from what remains a working law. The central outcome is that the global $\pm E$ organization is algebraically explained, while local hotspot prominence is statistically linked to excess constructive coherence among the virtual $P \rightarrow Q \rightarrow P$ pathways.

1. Frozen Phase 3.2 setting

Item	Frozen choice
Hilbert space	8 qubits, dimension 256
Hamiltonian	Frozen 11-term random-Pauli model inherited from Phase 3.1
Seeds	25042000-25042011
Eigenspaces	Complete degeneracy-closed P subspaces; 64 groups of dimension 4 for reference seed
Perturbation families	Dephasing Z/ZZ and transverse X/XX; five perturbation terms
Regularization	$g_{\text{eta}}(x)=x/(x^2+\text{eta}^2)$, $\text{eta}=1\text{e-}3$
Primary response	Frozen Phase-3.1 hotspot prominence

2. Global layer: the $\pm E$ structure is explained

The decisive Phase 3.2 symmetry result is the existence, for the reference frozen Hamiltonian, of a Pauli-string generator $\Gamma = \text{XIXXIIXX}$ with

$$\Gamma H \Gamma^\dagger = -H$$

and exact perturbation covariance

$$\Gamma V \Gamma^\dagger = s V, \quad s = -1 \text{ (dephasing)}, +1 \text{ (transverse)}$$

This induces the projector mapping $P_{-E} = \Gamma P_E \Gamma^\dagger$ and $Q_{-E} = \Gamma Q_E \Gamma^\dagger$. For the regularized second-order effective operator

$$\Sigma_{-E} = P V Q g_{\text{eta}}(E - QHQ) Q V P$$

the sign is fixed, not ambiguous:

$$\Sigma_{-E} = -\Gamma \Sigma_E \Gamma^\dagger$$

The v32.2 residuals for the required mapping chain were at approximately 10^{-12} or below, while the deliberately wrong plus-sign test produced residual 2.0. Therefore the equal Frobenius norms at $\pm E$ are a consequence of algebraic covariance rather than an empirical coincidence.

3. What v32.3-v32.7 ruled in and ruled out

Version	Test	Key result	Interpretation
v32.3	Reference-seed local mechanism scan	Σ -based descriptors correlated with prominence; transverse signal appeared strong.	Promising but seed-specific.
v32.4	Across-seed confirmation	Transverse real_sigma median Spearman +0.4222; real_sigma/gap only +0.1882.	Σ contains information, but no single scalar law.
v32.5	Low-dimensional multivariate LOSO	Transverse pooled LOSO rho +0.2991, R^2 +0.0693; positive fold rho in 12/12 seeds.	Reproducible ranking signal, weak amplitude prediction.
v32.6	Weighted Q-spectrum descriptors	Transverse pooled LOSO rho +0.2571, R^2 -0.0578.	Simple Q-density/gap descriptors do not supply the missing variable.
v32.7	Internal matrix shape of Σ	Transverse pooled LOSO rho +0.2846, R^2 -0.2352.	Static matrix anisotropy/rank is not sufficient.

4. v32.8-v32.9: interference/coherence emerges

The key conceptual shift was to stop treating Σ only as a final operator and instead decompose it into individual virtual Q-state contributions:

$$\Sigma_E = \Sigma_q T_q, \quad T_q = g_{\text{eta}}(E - E_q) P V |q\rangle\langle q| V P$$

This permits a dimensionless coherence factor

$$C_E = \|\Sigma_q T_q\|_F / \Sigma_q \|T_q\|_F$$

C_E is high when the virtual pathways reinforce one another and low when they cancel strongly. v32.8 improved the cross-seed model for both families, and v32.9 then removed the multivariate fit and tested C_E directly.

Family	Metric	Median Spearman	Consistency
Dephasing	C_E seed-median rho	+0.3088	11/12 positive
Transverse	C_E seed-median rho	+0.5423	11/12 positive
Transverse	positive/negative balance	+0.4987	12/12 positive
Transverse	scalar cancellation	-0.4987	12/12 negative

5. v32.10-v32.11: intervention against a norm-preserving null

To test whether the association was genuinely about alignment/cancellation rather than the magnitudes of the individual pathways, the signs of the T_q contributions were randomized independently while every individual ||T_q||_F was kept unchanged. This destroys constructive/destructive alignment but preserves pathway strength.

The strict v32.11 confirmation used 1000 norm-preserving sign randomizations per pair and seed-level permutation tests. Its primary endpoint was the coherence excess

$$H_E = C_E - \langle C_E \rangle_{\text{null}}$$

Family	Median seed rho	Positive seeds	Sign-test p	Pooled rho	Gate
Dephasing	+0.3292	11/12	0.00317	+0.3039	5/5 - STRONG
Transverse	+0.3713	10/12	0.01929	+0.3068	5/5 - STRONG

Interpretation: after the individual pathway norms are held fixed, the excess coherence above the randomized alignment null still tracks hotspot prominence. This is stronger evidence than a static correlation, although it remains a computational intervention rather than an analytic derivation.

6. v32.12: provisional local hotspot law

Phase 3.2 can now state a compact empirical local-law parameter:

$$H_E = C_E - \langle C_E \rangle_{\text{null}}$$
$$C_E = ||\Sigma_q T_q||_F / \Sigma_q ||T_q||_F$$

The tested statement is monotonic rather than an exact amplitude equation:

hotspot prominence(E) increases statistically as H_E increases

Family	Median seed rho(H_E)	Positive seeds	Sign-test p	Pooled rho	Median H_E
Dephasing	+0.3294	12/12	0.00024	+0.3029	+0.01710
Transverse	+0.3630	10/12	0.01929	+0.3054	+0.01974

The strict TOP/MIDDLE/BOTTOM ordering is not universal: it is monotone in 5/12 dephasing seeds and 4/12 transverse seeds. Therefore H_E is currently a statistically reproducible ranking parameter, not a deterministic per-hotspot law.

7. Current two-layer Phase 3.2 picture

GLOBAL LAYER

$$\Gamma H \Gamma^\dagger = -H \Rightarrow E \leftrightarrow -E \Rightarrow P_{-E} = \Gamma P_E \Gamma^\dagger \text{ and } \Sigma_{-E} = -\Gamma \Sigma_E \Gamma^\dagger$$

LOCAL LAYER

$$H_E = C_E - \langle C_E \rangle_{\text{null}}, \text{ prominence } \uparrow \text{ statistically with } H_E \uparrow$$

The first layer explains why ±E partners are organized symmetrically. The second layer is the current mechanism candidate for why some of those symmetry-related pairs become stronger hotspots than others.

8. Established, supported, and still open

Status	Statement
Established for reference symmetry chain	Exact Γ anticommutation; ±E projector mapping; fixed minus sign for Σ covariance; Frobenius-norm symmetry follows algebraically.
Strongly supported across 12 seeds	Interference/coherence descriptors track prominence; norm-preserving randomization leaves a significant coherence-excess signal; H_E passes the current cross-seed gates.
Not established	A closed analytic equation predicting the numerical prominence amplitude; universal strict TOP/MID/BOTTOM ordering; proof that the same explicit Γ string applies unchanged to every seed.
Next scientific target	Derive or constrain the mapping prominence = f(H_E, ...) from operator perturbation structure, or identify the minimal additional variable needed for amplitude prediction.

9. Checkpoint conclusion

Phase 3.2 has moved from an empirical observation of ±E organization to a coherent two-layer mechanism. The global layer is algebraic: an anticommuting symmetry organizes the spectrum and the effective second-order operator. The local layer is now represented by excess pathway coherence H_E: hotspots are statistically stronger where the actual virtual Q pathways add more coherently than in a norm-preserving randomized reference. This is sufficiently stable to freeze as a checkpoint, but not yet sufficient to claim a complete analytic hotspot law.